Slide Set #11

Inference with the Linear Model with Predermined Regressors

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Version: Saturday 20th February, 2021 (h16:45)

Notes

Preample on asymptotic testing procedures

Asymptotic testing procedures require in deep reconsideration of notions such as significance level, size, power, optimality, etc.

- Nominal size: $=$ desired level of $\alpha = \Pr\{\text{Type 1 Error}\}$
- Actual size/exact size: $\alpha_n = \Pr\{\text{Type 1 Error at given } n\}$

As minimal requirement we would like that asymptotically

$$\text{size distortion} = (\alpha_n - \alpha) \longrightarrow 0 \text{ as } n \to \infty$$

Asymptotic optimality for tests
loosely speaking, we require the best power for large n. Asymptotic power must be carefully defined.

Notes

Heteroscedasticity robust standard errors

We don't have $\mathrm{AVar}(\boldsymbol{b})$. However, $\widehat{\mathrm{AVar}\,(b_k)} \xrightarrow{\mathsf{p}} \mathrm{AVar}(b_k)$

Estimates for approximate standard errors

$$\mathrm{Se}(b_k) := \sqrt{\frac{1}{n}\widehat{\mathrm{AVar}\,(b_k)}} = \sqrt{\frac{1}{n}\left(\boldsymbol{S}_{\boldsymbol{xx}}^{-1}\hat{\boldsymbol{S}}\boldsymbol{S}_{\boldsymbol{xx}}^{-1}\right)_{[k,k]}}$$

Remarks:

- (B2) is consistent with conditionally heteroscedasticity, therefore $\mathrm{Se}(b_k)$ are "*robust*" to the presence of conditional heteroscedasticity
- $\mathrm{Se}(b_k)$ is called *heteroscedasticity robust standard error*, or *White's standard error*, or *heteroscedasticity consistent standard error*

Notes

Testing individual regression coefficients

Consider the null hypothesis: $\mathsf{H}_0 : \beta_k = \beta_k^*$

Under (B1)–(B5) + H_0 we have

$$\sqrt{n}(b_k - \beta_k^*) \xrightarrow{\mathsf{d}} \mathcal{N}(0, \mathrm{AVar}(b_k))$$

and by Slutzky's theorem (combining convergence modes)

$$t_k \;=\; \sqrt{n}\frac{b_k - \beta_k^*}{\sqrt{\widehat{\mathrm{AVar}\,(b_k)}}} \;=\; \frac{b_k - \beta_k^*}{\mathrm{Se}(b_k)} \xrightarrow{\mathsf{d}} \mathcal{N}(0,1)$$

Notes

Default test on individual coefficients

$$\begin{cases} \mathsf{H}_0 : \beta_k = 0 \\ \mathsf{H}_1 : \beta_k \neq 0 \end{cases}$$

Decision rule at significance level α:

$$\text{reject } \mathsf{H}_0 \text{ if } |t_k| > q_{1-\frac{\alpha}{2}}$$

where $q_{1-\frac{\alpha}{2}}$ is the $(1-\frac{\alpha}{2})$-quantile of the $\mathcal{N}(0,1)$

Confidence intervals on individual coefficients

Let $(1-\alpha)$ be the confidence level. The confidence interval for β_k is given by

$$b_k \ \pm \ q_{1-\frac{\alpha}{2}} \, \mathrm{Se}(b_k)$$

where $q_{1-\frac{\alpha}{2}}$ is the $(1-\frac{\alpha}{2})$-quantile of the $\mathcal{N}(0,1)$

Notes

Testing linear hypothesis

Now we consider a set of J linear restrictions

$$\mathsf{H}_0 : \boldsymbol{R}\boldsymbol{\beta} = \boldsymbol{r}$$

Where you specify some $(J \times K)$ matrix $\boldsymbol{R} = (R_{jk})$ and a $J-$dimensional column vector $\boldsymbol{r}$.

The formulation is general enough to formalize many useful relationships:

$$\begin{aligned} R_{11}\beta_1 + R_{12}\beta_2 + \cdots + R_{1K}\beta_K &= r_1 \\ R_{21}\beta_1 + R_{22}\beta_2 + \cdots + R_{2K}\beta_K &= r_2 \\ &\vdots \\ R_{J1}\beta_1 + R_{J2}\beta_2 + \cdots + R_{JK}\beta_K &= r_J \end{aligned}$$

Here we want to jointly test J linear relationships with a *Wald-type test*

Notes

Proposition (Wald test for linear hypothesis)

Assume (B1)–(B5), assume $\mathrm{H}_0 : \boldsymbol{R}\boldsymbol{\beta} - \boldsymbol{r} = \boldsymbol{0}$, where $\boldsymbol{R}$ is a $(J \times K)$ full rank matrix. Then for $n \to \infty$

$$W_n \;=\; n(\boldsymbol{R}\boldsymbol{b} - \boldsymbol{r})'\Big(\boldsymbol{R}\widehat{\mathrm{AVar}\,(\boldsymbol{b})}\boldsymbol{R}'\Big)^{-1}(\boldsymbol{R}\boldsymbol{b} - \boldsymbol{r}) \;\xrightarrow{\mathrm{d}}\; \chi^2(J)$$

Proof. Reorganize notation as follows:

$$\begin{aligned} \boldsymbol{c}_n &= \sqrt{n}(\boldsymbol{R}\boldsymbol{b} - \boldsymbol{r}) \\ \boldsymbol{Q}_n &= \boldsymbol{R}\widehat{\mathrm{AVar}\,(\boldsymbol{b})}\boldsymbol{R}' \\ W_n &= \boldsymbol{c}_n{}'\boldsymbol{Q}_n^{-1}\boldsymbol{c}_n \end{aligned}$$

$\mathrm{H}_0 \implies \boldsymbol{r} = \boldsymbol{R}\boldsymbol{\beta}$. Replace $\boldsymbol{r}$ into $\boldsymbol{c}_n$ and obtain $\boldsymbol{c}_n = \boldsymbol{R}\sqrt{n}(\boldsymbol{b} - \boldsymbol{\beta})$.

But for n large $\sqrt{n}(\boldsymbol{b} - \boldsymbol{\beta})\} \xrightarrow{\mathrm{d}} \mathcal{N}(\boldsymbol{0}, \mathrm{AVar}(\boldsymbol{b}))$

Notes

Therefore

$$\begin{aligned} \boldsymbol{c}_n = \boldsymbol{R}\sqrt{n}(\boldsymbol{b} - \boldsymbol{\beta}) &\xrightarrow{\mathrm{d}} \boldsymbol{R}\;\mathcal{N}(\boldsymbol{0}, \mathrm{AVar}(\boldsymbol{b})) \\ &\xrightarrow{\mathrm{d}} \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}\,\mathrm{AVar}(\boldsymbol{b})\boldsymbol{R}') = \boldsymbol{c} \end{aligned}$$

Since $\widehat{\mathrm{AVar}\,(\boldsymbol{b})}$ is consistent, then

$$\boldsymbol{Q}_n \;=\; \boldsymbol{R}\widehat{\mathrm{AVar}\,(\boldsymbol{b})}\boldsymbol{R}' \xrightarrow{\mathrm{p}} \boldsymbol{R}\,\mathrm{AVar}(\boldsymbol{b})\boldsymbol{R}' \;=\; \boldsymbol{Q}$$

$\boldsymbol{R}$ is full rank, $\mathrm{AVar}(\boldsymbol{b})$ is positive definite $\implies \boldsymbol{Q}$ invertible. Then

$$W_n \;=\; \boldsymbol{c}_n{}'\boldsymbol{Q}_n^{-1}\boldsymbol{c}_n \;\xrightarrow{\mathrm{d}}\; \boldsymbol{c}'\boldsymbol{Q}^{-1}\boldsymbol{c}$$

Note that $\boldsymbol{c} \in \mathbb{R}^J$ is normally distributed with $\mathrm{Var}[\boldsymbol{c}] = \boldsymbol{Q}$. It follows that $\boldsymbol{c}'\boldsymbol{Q}^{-1}\boldsymbol{c} \sim \chi^2(J)$ ■

Notes

Test

$$\begin{cases} \mathsf{H}_0 : \boldsymbol{R}\beta = \boldsymbol{r} \\ \mathsf{H}_1 : \boldsymbol{R}\beta \neq \boldsymbol{r} \end{cases}$$

Decision rule at significance level α:

$$\text{reject } \mathsf{H}_0 \text{ if } w > q_{1-\alpha}$$

where

- w is the observed value of the Wald's statistic W_n
- $q_{1-\alpha}$ is the $(1-\alpha)$-quantile of $\chi^2(J)$

Notes

Testing nonlinear hypothesis

Suppose the regression model is

$$Y_i = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \text{error}$$

and we want to test the joint null hypothesis

$$\mathsf{H}_0 : \ \beta_2^2 + \beta_3^2 = 1 \quad \text{and} \quad \log\left(\frac{\beta_4}{\beta_1}\right) = 10$$

The null hypothesis can be equivalently formulated as the following system of two equations

$$\begin{cases} \beta_2^2 + \beta_3^2 - 1 & = 0 \\ \log(\beta_4) - \log(\beta_1) - 10 & = 0 \end{cases}$$

Notes

Let $\boldsymbol{t} = (t_1, t_2, t_3, t_4)'$. Consider a non linear function $\boldsymbol{a} : \mathbb{R}^4 \longrightarrow \mathbb{R}^2$

$$\boldsymbol{a}(\boldsymbol{t}) = \begin{cases} t_2^2 + t_3^2 - 1 \\ \log(t_4) - \log(t_1) - 10 \end{cases}$$

The previous null hypothesis can be reformulated as

$$\mathsf{H}_0 : \boldsymbol{a}(\boldsymbol{\beta}) = \mathbf{0}$$

In general we may need to *jointly* test J nonlinear relationships involving the regression coefficients. Think about this as testing J nonlinear equations of the K-dimensional coefficient vector

Notes

Jacobian/gradient of $\boldsymbol{a}(\cdot)$

let $\boldsymbol{a} : \mathbb{R}^K \longrightarrow \mathbb{R}^J$, and let $\boldsymbol{A}(\boldsymbol{\beta})$ the first derivatives (Jacobian matrix) of $\boldsymbol{a}(\cdot)$ computed at $\boldsymbol{\beta}$. The matrix $\boldsymbol{A}(\boldsymbol{\beta})$ will be of dimension $J \times K$

Suppose that $\boldsymbol{b}$ is the usual *unrestricted* OLS estimator of $\boldsymbol{\beta}$

Constistency under the map $\boldsymbol{a}(\cdot)$

under regularity conditions on $\boldsymbol{a}(\cdot)$, $\boldsymbol{a}(\boldsymbol{b}) \xrightarrow{\mathsf{p}} \boldsymbol{a}(\boldsymbol{\beta})$ and $\boldsymbol{A}(\boldsymbol{b}) \xrightarrow{\mathsf{p}} \boldsymbol{A}(\boldsymbol{\beta})$

Wald-type test

construct a test statistics for H_0 that measures whether $\boldsymbol{a}(\boldsymbol{b})$ is sufficiently close to $\mathbf{0}$ to sustain the null hypothesis. As in previous cases we can't use the Euclidean distance to measures such a distance, because we can't exclude that the random variable $\boldsymbol{a}(\boldsymbol{b})$ has a spherical distribution

Notes

Proposition (Wald test for nonlinear hypothesis)

Assume (B1)–(B5), assume $\mathsf{H}_0 : \boldsymbol{a}(\boldsymbol{\beta}) = \mathbf{0}$, $\boldsymbol{a}(\cdot)$ has continuous first derivatives, and $A(\boldsymbol{\beta})$ is full rank. As $n \to \infty$

$$W_n \;=\; n\boldsymbol{a}(\boldsymbol{b})'\Big(\boldsymbol{A}(\boldsymbol{b})\widehat{\mathrm{AVar}\,(\boldsymbol{b})}\boldsymbol{A}(\boldsymbol{b})'\Big)^{-1}\boldsymbol{a}(\boldsymbol{b}) \;\xrightarrow{\mathsf{d}}\; \chi^2(J)$$

Proof. Under (B1)–(B5): $\sqrt{n}(\boldsymbol{b}-\boldsymbol{\beta}) \xrightarrow{\mathsf{d}} \mathcal{N}(\mathbf{0}, \mathrm{AVar}(\boldsymbol{b}))$.

Apply the delta method

$$\boldsymbol{c}_n \;=\; \sqrt{n}(\boldsymbol{a}(\boldsymbol{b}) - \boldsymbol{a}(\boldsymbol{\beta})) \;\xrightarrow{\mathsf{d}}\; \mathcal{N}(\mathbf{0}, \boldsymbol{A}(\boldsymbol{\beta})\,\mathrm{AVar}(\boldsymbol{b})\boldsymbol{A}(\boldsymbol{\beta})') \;=\; c$$

Under $\mathsf{H}_0 : \boldsymbol{a}(\boldsymbol{\beta}) = \mathbf{0}$, previous reduces to

$$\sqrt{n}\boldsymbol{a}(\boldsymbol{b}) \xrightarrow{\mathsf{d}} \boldsymbol{c}$$

$\boldsymbol{b} \xrightarrow{\mathsf{p}} \boldsymbol{\beta}$ and $\boldsymbol{A}(\cdot)$ exists, it is continuous, and its full rank at $\boldsymbol{\beta}$. Therefore $\boldsymbol{A}(\boldsymbol{b}) \xrightarrow{\mathsf{p}} \boldsymbol{A}(\boldsymbol{\beta})$

Notes

$\widehat{\mathrm{AVar}\,(\boldsymbol{b})} \xrightarrow{\mathsf{p}} \mathrm{AVar}(\boldsymbol{b})$, hence

$$\boldsymbol{A}(\boldsymbol{b})\widehat{\mathrm{AVar}\,(\boldsymbol{b})}\boldsymbol{A}(\boldsymbol{b})' \xrightarrow{\mathsf{p}} \boldsymbol{A}(\boldsymbol{\beta})\,\mathrm{AVar}(\boldsymbol{b})\boldsymbol{A}(\boldsymbol{\beta})' \;=\; \mathrm{Var}[\boldsymbol{c}]$$

$\mathrm{AVar}(\boldsymbol{b})$ is positive definite, and $\boldsymbol{A}(\boldsymbol{\beta})$ is full rank. This implies that $\mathrm{Var}[\boldsymbol{c}]$ is invertible. Thus

$$W_n \;=\; \sqrt{n}\boldsymbol{a}(\boldsymbol{b})'\Big(\boldsymbol{A}(\boldsymbol{b})\widehat{\mathrm{AVar}\,(\boldsymbol{b})}\boldsymbol{A}(\boldsymbol{b})'\Big)^{-1}\sqrt{n}\boldsymbol{a}(\boldsymbol{b}) \;\xrightarrow{\mathsf{d}}\; \boldsymbol{c}'(\mathrm{Var}[\boldsymbol{c}])^{-1}\boldsymbol{c}$$

Observe that $\boldsymbol{c} \sim \mathcal{N}(\mathbf{0}, \mathrm{Var}[\boldsymbol{c}])$ with $J \times J$ variance-covariance matrix $\mathrm{Var}[\boldsymbol{c}]$, therefore

$$\boldsymbol{c}'(\mathrm{Var}[\boldsymbol{c}])^{-1}\boldsymbol{c} \sim \chi^2(J)$$

which completes the proof ■

Notes

Test

$$\begin{cases} \mathsf{H}_0 : \boldsymbol{a}(\boldsymbol{\beta}) = \mathbf{0} \\ \mathsf{H}_1 : \boldsymbol{a}(\boldsymbol{\beta}) \neq \mathbf{0} \end{cases}$$

Decision rule at significance level α:

$$\text{reject } \mathsf{H}_0 \text{ if } w > q_{1-\alpha}$$

where

- w is the observed value of the Wald's statistic W
- $q_{1-\alpha}$ is the $(1-\alpha)$-quantile of $\chi^2(J)$

Notes

Conditional homoscedasticity

The stationarity assumption implies that $\mathrm{E}[\varepsilon_i^2]$ is constant but conditional homo/hetero-skdasticity is allowed.

Assumption (B7): conditional homoskedasticity

Assume $\mathrm{E}[\varepsilon_i^2 \,|\, \boldsymbol{x}_i] = \sigma^2$ for all $i = 1, 2, \ldots, n$.

Under (B7)

$$\begin{aligned} \boldsymbol{S} &= \mathrm{E}[\varepsilon_i^2 \boldsymbol{x}_i \boldsymbol{x}_i'] \\ &= \mathrm{E}[\mathrm{E}[\varepsilon_i^2 \boldsymbol{x}_i \boldsymbol{x}_i' \,|\, \boldsymbol{x}_i]] \text{ (iterated expectation)} \\ &= \mathrm{E}[\mathrm{E}[\varepsilon_i^2 \,|\, \boldsymbol{x}_i] \boldsymbol{x}_i \boldsymbol{x}_i'] \text{ (pull out what's known)} \\ &= \mathrm{E}[\sigma^2 \boldsymbol{x}_i \boldsymbol{x}_i'] \\ &= \sigma^2 \,\mathrm{E}[\boldsymbol{x}_i \boldsymbol{x}_i'] = \sigma^2 \boldsymbol{\Sigma}_{\boldsymbol{xx}} \end{aligned}$$

Notes

Recall:

- $\hat{\boldsymbol{S}} = \frac{1}{n}\sum_{i=1}^{n} e_i^2 \boldsymbol{x}_i \boldsymbol{x}_i'$ is consistent for $\boldsymbol{S} = \mathrm{E}[\varepsilon_i^2 \boldsymbol{x}_i \boldsymbol{x}_i']$
- $s^2 \boldsymbol{S}_{\boldsymbol{xx}} = \frac{1}{n}\sum_{i=1}^{n} s^2 \boldsymbol{x}_i \boldsymbol{x}_i'$ is consistent for $\sigma^2 \boldsymbol{\Sigma}_{\boldsymbol{xx}}$

Under (B7) we expect that the difference $(\hat{\boldsymbol{S}} - s^2 \boldsymbol{S}_{\boldsymbol{xx}})$ is close to zero in probability, that is

$$\begin{aligned}\hat{\boldsymbol{S}} - s^2 \boldsymbol{S}_{\boldsymbol{xx}} &= \frac{1}{n}\sum_{i=1}^{n} e_i^2 \boldsymbol{x}_i \boldsymbol{x}_i' - \frac{1}{n}\sum_{i=1}^{n} s^2 \boldsymbol{x}_i \boldsymbol{x}_i' \\ &= \frac{1}{n}\sum_{i=1}^{n} (e_i^2 - s^2) \boldsymbol{x}_i \boldsymbol{x}_i' \xrightarrow{\mathsf{p}} \boldsymbol{0}\end{aligned}$$

Notes

Conditional homoscedasticity and testing $\mathrm{H}_0 : \beta_k = \beta_k^*$

Consider the t-ratio statistic we computed in finite sample setup (fs):

$$t_k^{\mathsf{fs}} = \frac{b_k - \beta_k^*}{\sqrt{s^2 (\boldsymbol{X}'\boldsymbol{X})^{-1}_{[k,k]}}} = \frac{b_k - \beta_k^*}{\sqrt{\frac{1}{n} s^2 (\boldsymbol{S}_{\boldsymbol{xx}}^{-1})_{[k,k]}}}$$

And the one introduced in the large sample (ls) case

$$t_k^{\mathsf{ls}} = \frac{b_k - \beta_k^*}{\sqrt{\frac{1}{n} (\boldsymbol{S}_{\boldsymbol{xx}}^{-1} \hat{\boldsymbol{S}} \boldsymbol{S}_{\boldsymbol{xx}}^{-1})_{[k,k]}}}$$

Observe: $t_k^{\mathsf{fs}} = t_k^{\mathsf{ls}}$ when $\hat{\boldsymbol{S}} = s^2 \boldsymbol{S}_{\boldsymbol{xx}}$

Notes

Under (B7) the statistic t_k^{ls} is asymptotically equivalent to t_k^{fs} because in this case $\hat{\boldsymbol{S}} - s^2\boldsymbol{S}_{\boldsymbol{xx}} \xrightarrow{\mathsf{p}} 0$

Under (B1)–(B7) + $\mathsf{H}_0 : \beta_k = \beta_k^*$, $\quad t_k^{\mathsf{fs}} \xrightarrow{\mathsf{d}} \mathcal{N}(0,1)$ for n large enough

This means that regarding the default test ($\mathsf{H}_0 : \beta_k = 0$) we can still continue to use the finite sample formula *if we knew that (B7) hold...* but we never know, so if we have enough n we are safer using t_k^{ls}

Notes

Conditional homoscedasticity and testing $\mathsf{H}_0 : \boldsymbol{R\beta} = \boldsymbol{r}$

Let's go back to the linear restriction test: $\mathsf{H}_0 : \boldsymbol{R\beta} = \boldsymbol{r}$. Rewrite the F-statistic (finite sample case) using $(\boldsymbol{X}'\boldsymbol{X})^{-1} = \frac{1}{n}\boldsymbol{S}_{\boldsymbol{xx}}^{-1}$

$$F = \frac{(\boldsymbol{Rb} - \boldsymbol{r})'\left(\frac{1}{n}s^2\boldsymbol{R}\boldsymbol{S}_{\boldsymbol{xx}}^{-1}\boldsymbol{R}'\right)^{-1}(\boldsymbol{Rb} - \boldsymbol{r})}{J}$$

Compare it with the Wald statistic

$$W_n = n(\boldsymbol{Rb} - \boldsymbol{r})'\left(\boldsymbol{R}\boldsymbol{S}_{\boldsymbol{xx}}^{-1}\hat{\boldsymbol{S}}\boldsymbol{S}_{\boldsymbol{xx}}^{-1}\boldsymbol{R}'\right)^{-1}(\boldsymbol{Rb} - \boldsymbol{r})$$

Therefore if $\hat{\boldsymbol{S}} = s^2\boldsymbol{S}_{\boldsymbol{xx}}$, then $W_n = JF$. But, if (B1)–(B7) holds, then we expect that $\hat{\boldsymbol{S}} - s^2\boldsymbol{S}_{\boldsymbol{xx}} \xrightarrow{\mathsf{p}} 0$ and $W_n - JF \xrightarrow{\mathsf{p}} 0$

Concluding: under (B1)–(B7) + $\mathsf{H}_0 : \boldsymbol{R\beta} = \boldsymbol{r}$, $JF \xrightarrow{\mathsf{d}} \chi^2(J)$

Notes

White's test

We want to tests the null hypothesis that

$$\mathsf{H}_0 : \mathrm{E}[\varepsilon_i^2 \,|\, \boldsymbol{x}_i] = \text{ constant (conditional homoskedasticity)}$$

White's test statistic exploits the fact that under (B7) we expect

$$\frac{1}{n}\sum_{i=1}^{n}(e_i^2 - s^2)\boldsymbol{x}_i\boldsymbol{x}_i' \xrightarrow{\mathsf{p}} \mathbf{0}$$

Roughly speaking: under (B7) we expect that $(e_i^2 - s^2)$ is somewhat "orthogonal" to the elements of $\boldsymbol{x}_i\boldsymbol{x}_i'$.

Test procedure: check whether the squared errors linear functions of all the unique elements of $\boldsymbol{x}_i\boldsymbol{x}_i'$

Remark: unique elements of $\boldsymbol{x}_i\boldsymbol{x}_i'$ include $x_{i1}, x_{i2}, \ldots x_{iK}$, then $x_{i1}^2, x_{i2}^2, \ldots x_{iK}^2$, and all cross-products $x_{ij}x_{il}$

Notes

White's test procedure

1. Fit the original model
$$y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + \varepsilon_i$$
compute residuals e_i, and predicted values $\hat{y}_i$
2. Fit the auxiliary regression model:
$$e_i^2 = \theta_1 + \theta_2\hat{y}_i + \theta_3\hat{y}_i^2 + \text{error}$$
Remark: $\hat{y}_i$ and $\hat{y}_i^2$ are linear functions of the unique elements of $\boldsymbol{x}_i\boldsymbol{x}_i'$
3. Perform the auxiliary test
$$\begin{cases} \mathsf{H}_0^{\mathsf{aux}} : \theta_2 = \theta_3 = 0 \\ \mathsf{H}_1^{\mathsf{aux}} : \text{otherwise} \end{cases}$$
Rejection of $\mathsf{H}_0^{\mathsf{aux}}$ is equivalent to rejecting the original H_0

Notes

Testing for serial correlation

For the process $\{\varepsilon_t\}$ we want to be able to test:

$\mathsf{H}_0: \quad \rho_1 = \rho_2 = \ldots = \rho_p = 0$

$\mathsf{H}_1: \quad \rho_j \neq 0$ for at least a $j \in \{1, 2, \ldots, p\}$

Why? MDS has no serial correlation. Correlated residuals show evidence of a failure of (B5)

There are methods (e.g. the Durbin-Watson test) to test the null hypothesis of no serial correlation in the error term, however they require strictly exogenous regressors

Strict exogeneity assumption is unlikely to hold in time series models

Notes

Asymptotic distribution of sampling autocorrelations

Assume $\mathrm{E}[\varepsilon_t] = 0$, the sample autocorrelation function is

$$\tilde{\rho}_j = \frac{\tilde{\gamma}_j}{\tilde{\gamma}_0} \qquad \text{where} \quad \tilde{\gamma}_j = \frac{1}{n}\sum_{i=j+1}^{n} \varepsilon_i \varepsilon_{i-j}, \quad j = 0, 1, 2, \ldots$$

Fix p and define $\tilde{\boldsymbol{\rho}} = (\tilde{\rho}_1, \tilde{\rho}_2, \ldots, \tilde{\rho}_p)'$. If H_0 holds, then we expect that the *size/length* of $\boldsymbol{\rho}$ is close to zero

MDS + "own" conditional homoscedasticity

suppose $\{\varepsilon_i\}$ is a stationary MDS and

$$\mathrm{E}[\varepsilon_i^2 \,|\, \varepsilon_{i-1}, \varepsilon_{i-2}, \ldots] = \sigma^2$$

Key result (see Hall and Heyde, 1980)

$$\sqrt{n}\tilde{\boldsymbol{\rho}} \xrightarrow{\mathsf{d}} \mathcal{N}(\mathbf{0}, \boldsymbol{I}_p)$$

Since $\sqrt{n}\tilde{\boldsymbol{\rho}}$ is asymptotically spherically normal, we can test H_0 by checking that $\|\boldsymbol{\rho}\|_2^2$ is sufficiently small

Notes

Box-Pierce and Ljung-Box test

Box-Pierce statistic

$$Q_{BP} = \sum_{j=1}^{p} (\sqrt{n}\tilde{\rho}_j)^2 = n \sum_{j=1}^{p} \tilde{\rho}_j^2 \xrightarrow{\mathsf{d}} \chi^2(p)$$

It is small when H_0 holds

Ljung-Box statistic

$$Q_{LB} = n(n+2) \sum_{j=1}^{p} \frac{\tilde{\rho}_j^2}{n-j} \xrightarrow{\mathsf{d}} \chi^2(p)$$

This is also small when H_0 holds

$Q_{LB} \overset{a}{\sim} Q_{BP}$, but for smaller n the Ljung-Box Q_{LB} provides better convergence to the previous χ^2 limit

Notes

Remark 1 ⚠
distributions of both Q_{BP} and Q_{LB} are derived under stationary MDS + own conditional homoscesticity

Remark 2
p is a tuning, how large should it be taken? There is some rule of thumb... not rocket science. Usually use $p \approx 10$ for non-seasonal data, and $p \approx (2\times\text{period})$ for seasonal data.

Remark 3 ⚠
we don't know $\{\varepsilon_i\}$, in fact we can only reconstruct it by $\{e_i\}$. Does $\{e_i\}$ do a good job?

- **with strictly exogenous regressors:** Q_{BP} and Q_{LB} computed over the residual $\{e_i\}$ still have the previous $\chi^2(p)$ limit.
- **with predetermined regressors:** ... we don't know! If have lagged regressors (time series data), it's better to stay away from using Q_{BP} and Q_{LB}.

Notes

Modified Box-Pierce Q

Assumption (Strong predeterminedness/conditional homoskedasticity)

The error process $\{\varepsilon_i\}$ is such that for all $i = 1, 2, \ldots, n$

(B5-ERR): $\mathrm{E}[\varepsilon_i | \varepsilon_{i-1}, \varepsilon_{i-2}, \ldots, \boldsymbol{x}_i, \boldsymbol{x}_{i-1}, \ldots] = 0$

(B7-ERR): $\mathrm{E}[\varepsilon_i^2 | \varepsilon_{i-1}, \varepsilon_{i-2}, \ldots, \boldsymbol{x}_i, \boldsymbol{x}_{i-1}, \ldots] = \sigma^2$

Remarks:

- (B5-ERR) is a stronger form of predeterminedness (homeworks)
- (B7-ERR) is a stronger form of the conditional homoscedasticity formalized in (B7)

Let $\hat{\boldsymbol{\rho}} = (\hat{\rho}_1, \hat{\rho}_2, \ldots, \hat{\rho}_p)'$ be sample correlations computed on residuals $e_1, e_2, \ldots, e_n$ obtained from the OLS estimation.

Notes

Proposition (Autocorrelation test statistic based on residuals)

Assume (B1), (B2), (B4), (B5-ERR), (B7-ERR). Then as $n \to \infty$

$$\sqrt{n}\hat{\boldsymbol{\rho}} \xrightarrow{\mathsf{d}} \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_p - \boldsymbol{\Phi})$$

where $\boldsymbol{\Phi}_{[jk]} = \sigma^{-2}\,\mathrm{E}(\boldsymbol{x}_i\varepsilon_{i-j})'\,\mathrm{E}(\boldsymbol{x}_i\boldsymbol{x}_i')^{-1}\,\mathrm{E}(\boldsymbol{x}_i\varepsilon_{i-k})$

Consistent estimation of $\hat{\boldsymbol{\Phi}}$ is performed by

$$\hat{\boldsymbol{\Phi}}_{[jk]} = s^{-2}\left(\frac{1}{n}\sum_{i=j+1}^{n}\boldsymbol{x}_i e_{i-j}\right)'\boldsymbol{S}_{\boldsymbol{xx}}^{-1}\left(\frac{1}{n}\sum_{i=j+1}^{n}\boldsymbol{x}_i e_{i-j}\right)$$

Modified Box-Pierce test statistic:

$$Q_{MBP} = n\,\hat{\boldsymbol{\rho}}'\left(\boldsymbol{I}_p - \hat{\boldsymbol{\Phi}}\right)^{-1}\hat{\boldsymbol{\rho}} \xrightarrow{\mathsf{d}} \chi^2(p)$$

Q_{MBP} is small when H_0 holds

Notes

Breusch-Godfrey test

Testing procedure based on a statistic asymptotically equivalent to Q_{MBP}.

Testing procedure:

1. Estimate the original model $y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + \varepsilon_i$, and get the residuals $\{e_i\}$
2. Estimate the auxiliary regression model

$$\begin{aligned} e_i = \phi_1 x_{i1} + \phi_2 x_{i2} +, \ldots \phi_K x_{iK} \\ + \theta_1 e_{i-1} + \theta_2 e_{i-2} +, \ldots, + \theta_{i-p} e_{i-p} + \text{error} \end{aligned}$$

   let R^2_{aux} be the R^2 statistic from the this auxiliary regression.
   Breusch-Godfrey test statistics

$$(n-p)R^2_{\text{aux}} \xrightarrow{\text{d}} \chi^2(p)$$

   that is asymptotically equivalent to Q_{MBP}.
3. Reject H_0 when $(n-p)R^2_{\text{aux}}$ exceeds the $(1-\alpha)$ quantile of $\chi^2(p)$

Notes

Notes